

The mark cost is exactly $\hbar/2$, and the worst-case theorem survives as a statistical one

Reopening the question of what was wrong with the derivation note: almost nothing. Its Proposition 6 claimed a positive quantum value for the mark cost $\kappa = \delta\Delta$, and that claim is correct. The proof was invalid and the constant was too large by a factor of four, and the worst-case *formulation* of δ and Δ as support widths is what made the quantum instance empty. Replacing support widths by standard deviations repairs everything at once, and two theorems then close the gap the derivation note left open.

Theorem A. For a mark realized by the impulsive coupling $\lambda\hat{y} \otimes \hat{P}_A$ of the body's position to a probe, with the probe's pointer read afterwards, the mark's error operator and the impulse it delivers are canonically conjugate,

$$\hat{N} = \frac{\hat{Q}_A}{\lambda} - \hat{y}, \quad \hat{D} = -\lambda\hat{P}_A, \quad [\hat{N}, \hat{D}] = -i\hbar, \quad \text{hence} \quad \delta\Delta = \Delta_\psi \hat{N} \cdot \Delta_\psi \hat{D} \geq \frac{\hbar}{2}$$

for every probe state whatever. So $\kappa \geq \hbar/2$, exactly, with no error–disturbance relation and no calibration argument.

Theorem B. In the Gaussian statistical model, for every protocol of marks with $\delta_j\Delta_j \geq \kappa$, of any number and at any times, with the initial position and velocity unknown, the optimal test's deflection obeys

$$d^2 \leq \frac{2F^2\tau^3}{9m\kappa}, \quad \text{hence deciding at error probability } \epsilon \text{ requires} \quad \tau\Delta E = \frac{F^2\tau^3}{2m} \geq 9z_{1-\epsilon}^2\kappa,$$

with $z_{1-\epsilon} = \Phi^{-1}(1-\epsilon)$. With Theorem A this is $\tau\Delta E \geq \frac{9}{2}z_{1-\epsilon}^2\hbar$, about $12\hbar$ at five per cent error. The proof is the derivation note's proof unchanged: the same certificate, the same arithmetic–geometric step, the same integration by parts to a Dirichlet energy, with the separating hyperplane replaced by the optimal test direction. The constant 9 reappears from the same Cauchy–Schwarz inequality.

Newton's premise M3 is Theorem A for his own probe: the interval of fits Λ is the corpuscle's transverse position spread and p its transverse momentum spread, so $\kappa = \Lambda p$ is the corpuscle's own uncertainty product, and the heuristic value $\hbar/2$ of the mark-floor note differs from the theorem's $\hbar/2$ by π . Section 6 sets out the single paper this supports. Exploratory; no ledger promotion.

1. What was actually wrong, stated exactly

The derivation note's framework defines a mark's resolution δ and recoil width Δ as worst-case interval widths: the datum confines the position to an interval of length δ with certainty, and the records confine the delivered impulse to an interval of length Δ with certainty. Three claims about that framework are now settled.

- **Theorem 1 is correct and is not affected.** For every finite protocol with $\delta_j\Delta_j \geq \kappa$, deciding the comparison needs $F^2\tau^3 > 9m\kappa$.
- **The worst-case framework has no quantum instance.** Certain position confinement forces a compactly supported conditional state, whose momentum distribution has full support by Paley–Wiener. So no quantum mark has both widths finite, $\kappa = \infty$, and Theorem 1 holds empty. This is the correction that stands.
- **The conclusion of the first Proposition 6 was right anyway.** Its assertion was that quantum kinematics gives κ a positive value of the order of $\hbar$, and that the Galileo comparison therefore has a floor of order $\hbar$ in $\tau\Delta E$. Theorem A gives $\kappa \geq \hbar/2$ and Theorem

B gives the floor, so both are true. What failed was the route: bounding a Wasserstein-2 calibration disturbance by half a worst-case width, when that disturbance is infinite for the instruments in question.

So the defect was the choice of currency, and it cost one constant and one proof. Standard deviations are the right currency because they are finite for quantum states, because they are what the uncertainty relation constrains, and because the derivation note's proof never used containment beyond a quadratic bound, as Section 3 shows.

2. Theorem A: the mark's error and its recoil are conjugate

Take the von Neumann model of a position mark. The body has $\hat{y}, \hat{p}$; the probe has $\hat{Q}_A, \hat{P}_A$ with $[\hat{Q}_A, \hat{P}_A] = i\hbar$; the mark is the impulsive unitary

$$U = \exp\left(-\frac{i}{\hbar}\lambda\hat{y}\hat{P}_A\right), \quad \lambda > 0.$$

Theorem A. In the Heisenberg picture across the mark,

$$U^\dagger \hat{Q}_A U = \hat{Q}_A + \lambda\hat{y}, \quad U^\dagger \hat{p} U = \hat{p} - \lambda\hat{P}_A, \quad U^\dagger \hat{y} U = \hat{y}, \quad U^\dagger \hat{P}_A U = \hat{P}_A.$$

Reading the pointer $\hat{Q}_A$ and estimating the body's position by $\hat{Q}_A/\lambda$, the error operator $\hat{N} = \hat{Q}_A/\lambda - \hat{y}$ and the delivered impulse $\hat{D} = \hat{p}_{\text{after}} - \hat{p}_{\text{before}} = -\lambda\hat{P}_A$ satisfy $[\hat{N}, \hat{D}] = -i\hbar$, hence for every joint state

$$\delta\Delta := \Delta\hat{N} \cdot \Delta\hat{D} \geq \frac{\hbar}{2}.$$

Proof. With $A = \frac{i}{\hbar}\lambda\hat{y}\hat{P}_A$, so that $U^\dagger = e^A$, the commutators are $[A, \hat{Q}_A] = \frac{i\lambda\hat{y}}{\hbar}[\hat{P}_A, \hat{Q}_A] = \lambda\hat{y}$ and $[A, \hat{p}] = \frac{i\lambda\hat{P}_A}{\hbar}[\hat{y}, \hat{p}] = -\lambda\hat{P}_A$, both central, so the Baker–Campbell–Hausdorff series terminates and gives the four displayed relations. The error operator is then $\hat{N} = (\hat{Q}_A + \lambda\hat{y})/\lambda - \hat{y} = \hat{Q}_A/\lambda$, whence

$$[\hat{N}, \hat{D}] = \left[\frac{\hat{Q}_A}{\lambda}, -\lambda\hat{P}_A\right] = -[\hat{Q}_A, \hat{P}_A] = -i\hbar,$$

and Robertson's inequality gives $\Delta\hat{N} \Delta\hat{D} \geq \frac{1}{2}|\langle[\hat{N}, \hat{D}]\rangle| = \hbar/2$. $\square$

Three features matter. The coupling strength λ cancels, so the bound is a property of the mark and not of how hard it is made. The result holds for every probe state, pure or mixed, Gaussian or not. And the two quantities are the probe's own position and momentum spreads: $\delta = \Delta\hat{Q}_A/\lambda$ and $\Delta = \lambda\Delta\hat{P}_A$, so

$$\kappa = \delta\Delta = \Delta\hat{Q}_A \cdot \Delta\hat{P}_A,$$

the mark's cost is the probe's uncertainty product, and the floor of the Galileo comparison is set by the probe rather than by the body. The contested error–disturbance relations play no part: Ozawa's counterexamples concern instruments whose commutator $[\hat{N}, \hat{D}]$ differs from $-i\hbar$, and the present statement is confined to the momentum-transfer class where it equals $-i\hbar$.

Why the records cannot recover the recoil. The pointer record is $\hat{R} = \hat{Q}_A + \lambda\hat{y}$ and the impulse is $\hat{D} = -\lambda\hat{P}_A$, with $[\hat{R}, \hat{D}] = -i\hbar\lambda$. The record and the recoil are incompatible observables, so no reading of the pointer, however precise, determines the impulse. This is Proposition 5 of the derivation note seen from the other side: the far-screen construction recovers the impulse by measuring the probe's momentum, and measuring the probe's momentum is exactly what reading its position forbids.

3. Theorem B: the same proof, in the Gaussian statistical model

The model. Marks at $0 = t_0 < t_1 < \dots < t_k \leq \tau$. Mark j returns $R_j = y(t_j) + \xi_j$ with ξ_j centred Gaussian of standard deviation δ_j , and delivers an impulse ι_j , centred Gaussian of standard deviation Δ_j , all independent. The trajectory is

$$y(t) = y_0 + v_0 t + \theta P(t) + \frac{1}{m} \sum_j \iota_j (t - t_j)_+, \quad P(t) = \frac{F t^2}{2m},$$

with $\theta = 0$ under I and $\theta = 1$ under F, and y_0, v_0 unknown. A protocol satisfies the **mark trade-off** with constant κ when $\delta_j \Delta_j \geq \kappa$ for every j .

Invariance and the deflection. Tests are required to be invariant under the unknown y_0 and v_0 , so the statistic is $u^\top R$ with u orthogonal to the vectors $(1, \dots, 1)$ and $(t_0, \dots, t_k)$. For Gaussian data the optimal invariant test has error probability $\Phi(-d/2)$ with

$$d^2 = \sup \left\{ \frac{(u^\top P)^2}{u^\top \Sigma u} : \sum_i u_i = 0, \sum_i u_i t_i = 0 \right\}, \quad P_i = P(t_i),$$

Σ being the covariance of R . This u is the statistical counterpart of the derivation note's separating multipliers μ , and it obeys the same constraint $\sum_i u_i = 0$.

Theorem B. For every such protocol, $d^2 \leq \frac{2F^2\tau^3}{9m\kappa}$. Consequently an invariant test with error probability at most ϵ requires

$$F^2\tau^3 \geq 18z_{1-\epsilon}^2 m\kappa, \quad \tau\Delta E \geq 9z_{1-\epsilon}^2 \kappa, \quad A_{\text{inertial,fall}} \geq \frac{6z_{1-\epsilon}^2 v\kappa}{F}.$$

Proof. The noise decomposes exactly as in the derivation note's feasibility system: ξ_j enters R_j alone, and ι_j enters R_i for $i > j$ with coefficient $(t_i - t_j)/m$. Hence

$$u^\top \Sigma u = \sum_j \delta_j^2 u_j^2 + \sum_j \frac{\Delta_j^2}{m^2} S_j^2, \quad S_j = \sum_{i>j} u_i (t_i - t_j),$$

the same S_j as there. By the arithmetic–geometric mean inequality applied termwise, exactly as in that note's Lemma 2,

$$u^\top \Sigma u \geq \sum_j \frac{2\delta_j \Delta_j}{m} |u_j| |S_j| \geq \frac{2\kappa}{m} \sum_j |u_j| |S_j|.$$

Since $\sum_i u_i = 0$, Lemma 3 of that note applies verbatim with μ replaced by u : with $T(w) = \int_w^\tau \sum_{i:t_i > s} u_i ds$ one has $T(t_j) = S_j$,

$$u^\top P = \frac{F}{m} \int_0^\tau T, \quad \sum_j u_j T(t_j) = -E, \quad E = \int_0^\tau T'^2,$$

so $\sum_j |u_j| |S_j| \geq E$ and $u^\top \Sigma u \geq 2\kappa E/m$. Cauchy–Schwarz with $T(\tau) = 0$ gives $\int_0^\tau T \leq \frac{2}{3}\tau^{3/2}\sqrt{E}$, so

$$\frac{(u^\top P)^2}{u^\top \Sigma u} \leq \frac{(F/m)^2 \frac{4}{9} \tau^3 E}{2\kappa E/m} = \frac{2F^2\tau^3}{9m\kappa},$$

independently of u , of the number of marks and of their times. The error probability $\Phi(-d/2) \leq \epsilon$ requires $d \geq 2z_{1-\epsilon}$, and $F^2\tau^3 \geq \frac{9}{2}m\kappa d^2 \geq 18z_{1-\epsilon}^2 m\kappa$. $\square$

The worst-case theorem and the statistical theorem therefore have one proof. The certificate changes meaning, from a hyperplane separating a feasible set to the direction of the optimal test, and every step after it is identical. This is the sense in which the derivation note's content survived its formulation: the argument never needed interval containment, only a quadratic bound in the same two variables.

4. Why Gaussian quantum experiments are covered exactly

Within its class the statistical model of Section 3 reproduces the quantum one exactly. For Gaussian probe states, linear dynamics and pointer readouts, the Wigner function is a genuine probability density and evolves by the classical linear equations, so every measured distribution in the quantum experiment equals the corresponding distribution in the classical Gaussian model whose noise covariance is the probe's Wigner covariance. Theorem A identifies that covariance's uncertainty product as $\delta\Delta \geq \hbar/2$. Hence for every protocol of Gaussian momentum-transfer marks on a body under the two hypotheses,

$$\tau\Delta E \geq \frac{9}{2} z_{1-\epsilon}^2 \hbar.$$

At $\epsilon = 0.05$, where $z = 1.645$, this is $\tau\Delta E \geq 12.2\hbar$; at $\epsilon = 0.32$, where $z = 1$, it is $4.5\hbar$. The insertion statement follows as before: marks inserted into a window of duration τ' , used by themselves, record the force only if $F^2\tau'^3 \geq 18z^2m\kappa$, so Democritus insertion stops at the mesh

$$\tau_* = \left(\frac{18 z_{1-\epsilon}^2 m \kappa}{F^2} \right)^{1/3} = \left(\frac{9 z_{1-\epsilon}^2 m \hbar}{F^2} \right)^{1/3},$$

and finer marks record free motion at the stated confidence.

This coexists with the counterexample of the probabilistic note, which distinguishes the hypotheses at arbitrarily small $\tau\Delta E$ using a two-packet preparation of separation $\pi\hbar/(F\tau)$. That protocol is non-Gaussian and makes no marks: it prepares once, waits, and measures once. The two results divide the ground cleanly. **Marking the trajectory costs $\hbar/2$ per mark and gives a floor of order $\hbar$; declining to mark it costs an apparatus of size $\pi\hbar/(F\tau)$ and gives the floor $\hbar^2/(4A)$.** Newton's refinement is the first case, which is why the floor is the relevant statement for the insertion question.

5. Newton's premise is Robertson's inequality for his corpuscle

The mark-floor note posited M3, that a mark fixing the position within δ leaves the delivered impulse undetermined within $\Lambda p/\delta$, and observed that Newton's optics supplies Λ and p while his determinism denies the premise. Theorem A says what M3 is. The corpuscle is the probe; $\hat{Q}_A$ is its transverse position and $\hat{P}_A$ its transverse momentum; the interval of fits Λ is the scale of the first and the corpuscular impulse p the scale of the second; and the premise asserts precisely that their product cannot be reduced. So

$$\kappa = \Lambda p \quad \longleftrightarrow \quad \kappa = \Delta\hat{Q}_A \cdot \Delta\hat{P}_A \geq \frac{\hbar}{2},$$

and M3 is Robertson's inequality for the corpuscle, stated in the two quantities Newton had. Newton measured the first as $1/89000$ inch, had no access to the second, held both to be determinate properties of the corpuscle, and thereby denied exactly the inequality. The modern value $\hbar/2$ against the identification $\Lambda p = \hbar/2$ differ by π .

The historical claim this licenses is narrow and checkable. Newton's optics contains both factors of the product that bounds the recorded Galileo comparison; his mathematics contains the comparison and takes it to zero; and the single proposition that would join them is a claim of indeterminacy about the fits, which he considered and rejected in favour of a determinate periodic disposition. The gap between the *Principia*'s vanishing sagitta and the *Opticks*' finite interval of fits follows from that single commitment, and the separation of the two books is incidental to it.

6. The single paper, valid for both audiences

The user's requirement is one paper that a foundations-of-physics referee and a history-and-philosophy-of-science referee both accept. That is attainable, because the historical analysis now does argumentative work rather than decorating a theorem. The structure is:

1. **The comparison and its limit.** Newton's Lemmas X and XI and the projectile Scholium, with the sagitta, the inertial-parabola area and the identity $(3F/v)A = \tau\Delta E$. Berkeley's objection and its resolution inside the geometry, with Guicciardini on what the limit arguments were for.
2. **The two theorems.** Theorem A, that a momentum-transfer mark's error and recoil are conjugate, and Theorem B, that every protocol of such marks needs $\tau\Delta E \geq 9z^2\kappa$. Proof by certificate, arithmetic-geometric mean and Dirichlet energy. This is the foundations content and it stands alone.
3. **The two factors in the *Opticks*.** The interval of fits as a measured length, the corpuscular impulse as a posited one, and Shapiro's account of what the theory of fits was and how determinate Newton meant it. This is the history content and it stands alone.
4. **The junction.** M3 is Robertson for the corpuscle. The counterfactual is replaced by an exact statement: the premise that the theorem needs is one Newton formulated the negation of, so the distance between his system and a positive h is a single proposition, nameable in his own vocabulary.
5. **What the theorem does not give.** The aperture counterexample, the standard-quantum-limit literature and Ozawa's class, and the honest prior-art label: the combination $F^2\tau^3 \gtrsim m\hbar$ is the standard quantum limit, and the contribution is universality over protocols, the exact mark cost, and the identification of the premise.

The venues that take this shape are Foundations of Physics and Studies in History and Philosophy of Modern Physics, both of which publish papers whose historical and technical halves are load-bearing for each other. Two obligations remain before submission: reading Shapiro, Guicciardini and Sabra rather than citing them, and replacing the Project Gutenberg *Opticks* and the Wikisource *Principia* with the fourth edition and the Cohen–Whitman translation.

7. Consequence for STATE

The derivation note's Proposition 6 is reinstated in substance with a correct proof and the constant $\hbar/2$ in place of $2\hbar$, and its worst-case formulation is replaced by standard deviations. The Planck gap for marked trajectories is $\tau\Delta E \geq \frac{9}{2}z_{1-\epsilon}^2\hbar$, protocol-universal, with the insertion mesh $\tau_* = (9z^2m\hbar/F^2)^{1/3}$. The remaining programme is item 6's two obligations plus: extending Theorem A beyond the momentum-transfer class, where Ozawa's instruments live; the general force law, which Theorem B's proof already reduces to a norm of P'' ; and closing the constants between the worst-case, statistical and single-shot forms.